

Cryptology Assignment Lab 2

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Importance of Cryptography: Using Wireshark, investigate the transmission of dummy login credentials over HTTP and HTTPS. Analyze the captured network traffic, compare how the credentials are transmitted in both protocols, and explain the role of encryption in protecting sensitive information.

Introduction

Cryptography is essential for protecting sensitive information as it travels across a network. Login credentials are particularly important because usernames and passwords can be used to access personal accounts, business systems, and other confidential resources. This experiment uses Wireshark to investigate the transmission of dummy login credentials over HTTP and HTTPS and compares the visibility of the captured traffic.

Objective

The objective of this experiment is to use Wireshark to capture and analyze network traffic generated when dummy login credentials are transmitted over **HTTP** and **HTTPS**, compare the visibility of the credentials in both protocols, and understand how encryption provided by HTTPS/TLS protects sensitive information from unauthorized interception. The main objective is to demonstrate how encryption protects credentials from being read by a network traffic monitor.

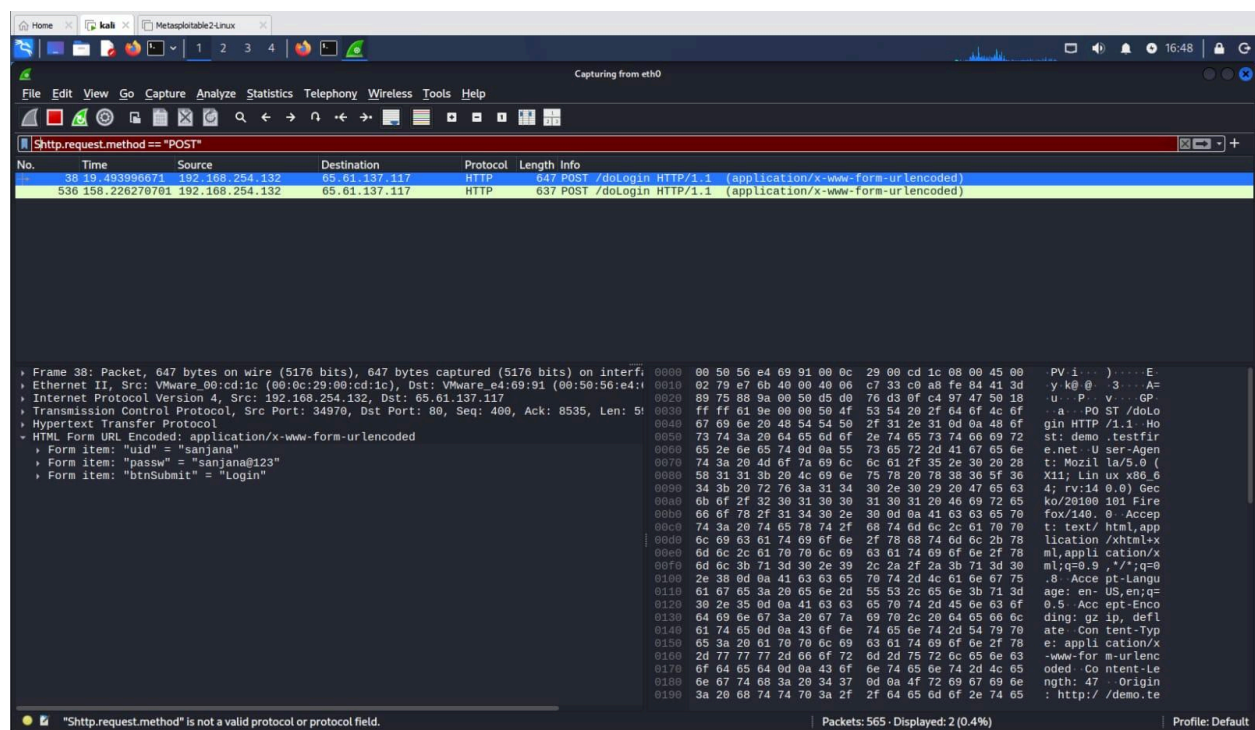
Experimental Setup

A test environment was used with dummy credentials rather than real account information. A login request was transmitted first through an HTTP connection and then through an HTTPS connection. Wireshark was used to capture the packets generated during both login attempts.

HTTP Traffic Findings

HTTP does not provide encryption for application-layer data. When a username and password are submitted through an HTTP login form, the request contents are transmitted in plaintext unless the application performs additional encryption itself.

In the HTTP capture, Wireshark shows the login information being sent as a **POST request** to the website. The username and password can be seen clearly in the captured data. This happens because HTTP does not hide the information being sent between the browser and the website. As a result, someone monitoring the network could potentially see the login details.



The screenshot also shows information such as the website address, the type of request, and the form details submitted by the user. This makes it easy to understand what information is being sent.

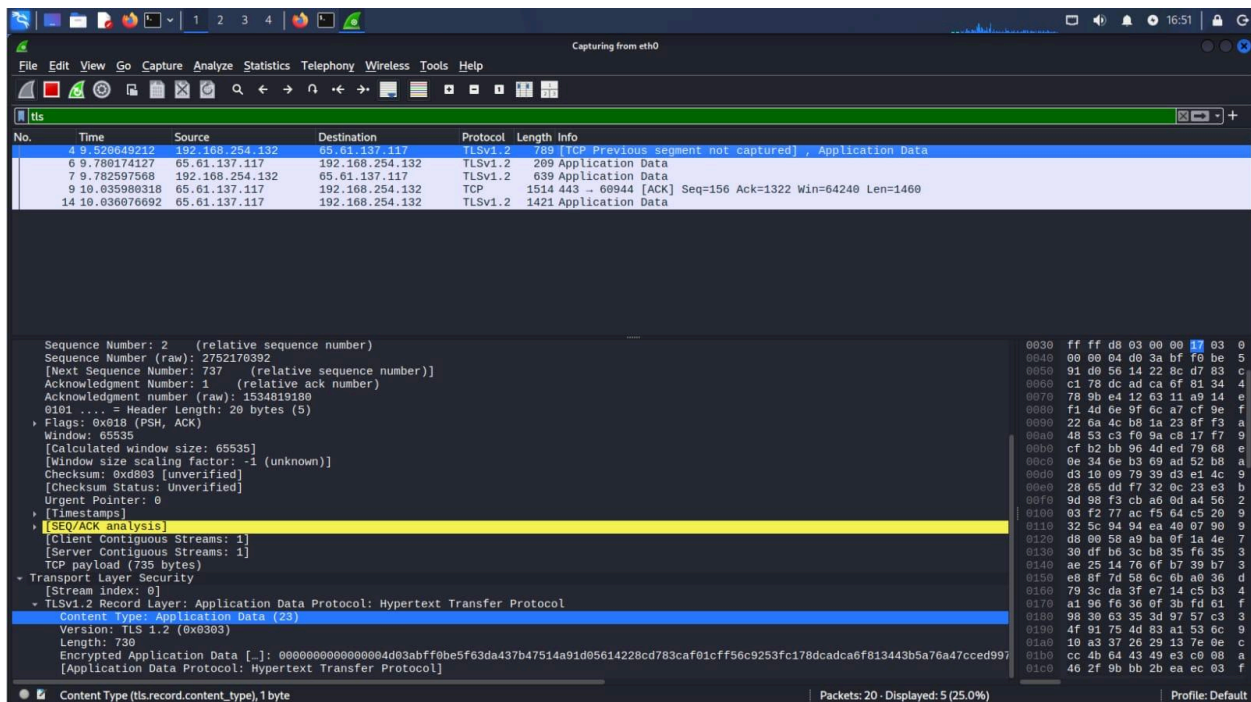
Most importantly, Wireshark has decoded the form data and shows:

- **Form item:** "uid" = "sanjana"
- **Form item:** "passw" = "sanjana@123"
- **Form item:** "btnSubmit" = "Login"

HTTPS Traffic Findings

HTTPS is HTTP transmitted through a TLS-protected connection. Before application data is exchanged, the client and server perform a TLS handshake to establish cryptographic parameters and session keys.

Because the information is encrypted, the username and password cannot be easily read from the captured packets. Instead, Wireshark shows a large amount of unreadable data. This means that even if someone captures the traffic, they cannot simply look at the packets and see the login credentials.



The important part is near the bottom: Encrypted Application Data

This means that the application data being sent between the client and server has been encrypted using TLS. Wireshark can see that data is being transmitted, but the actual HTTP contents, including any login credentials, are not displayed as readable text.

Comparison between the protocols

HTTP	HTTPS
Shows Encrypted Application Data	Shows readable POST request
Credentials are not directly visible	Username and password are clearly visible
Provides confidentiality	No encryption of the HTTP payload
Safer for login credentials	Vulnerable to network interception

Conclusion

The experiment demonstrates the importance of cryptography in network security. HTTP transmits login credentials without protecting their confidentiality, making them vulnerable to interception and inspection. HTTPS uses TLS encryption to protect the contents of the communication, preventing captured packets from directly revealing the transmitted credentials. Thus this proves that HTTPS is much safer for websites that handle passwords and other private information.